

Production Optimization and Food Waste Reduction Analysis Report

Case Study: De Gouden Tarwe Bakery (Netherlands)

Python — Machine Learning — DuckDB

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1 Project Overview

This project aims to design a data-driven analytical system to minimize losses due to unsold dough and bread leftovers (*food waste*) across the **De Gouden Tarwe** bakery network in the Netherlands. Fresh retail food products have an extremely short shelf-life (often only 1 day), making the imbalance between the quantity of dough produced and the quantity successfully sold a direct driver of poor business profitability.

Through the integration of a relational database, time-series feature engineering, and predictive modeling, this project identifies operational failure patterns, determines the branches with the highest waste costs, and compares the performance of machine learning models to generate optimal daily production quantity recommendations.

1.1 Business Problem

- **Massive Financial Loss:** Total operational losses due to food waste in the database reach **€3,404,818.03** (approximately IDR 59 Billion).
- **Inaccurate Production Estimations:** Current estimations of daily dough quantities per branch are done manually/conventionally, resulting in frequent overproduction or stockouts before the end of the day.
- **Sustainability Impact:** Wasting raw ingredients (flour, sugar, butter) causes indirect losses to supply chain logistics and negatively impacts the company's sustainability reputation.

1.2 Project Goals

1. Perform data cleaning, join relational tables, and engineer time-series features based on rolling windows and lags efficiently using the **DuckDB** database engine.
2. Analyze the primary causes of product disposal (*waste reason*) and map the branches with the highest financial waste losses.
3. Generate Return on Investment (ROI) estimates for raw material cost savings using Machine Learning methods to support business decision-making.
4. Correlation analysis of actual production vs sales and waste.

2 Literature Review of Prediction Models & Evaluation Metrics

2.1 XGBoost Regressor

Extreme Gradient Boosting (XGBoost) is a decision-tree-based ensemble learning algorithm that works by sequentially minimizing a loss function [1]. XGBoost Regressor is widely used to solve complex regression tasks and time-series forecasting challenges in industry [2]. The t -th tree is trained to predict the residuals of the $(t - 1)$ -th tree.

The objective function of XGBoost at iteration t is defined as:

$$\mathcal{L}^{(t)} = \sum_{i=1}^n l\left(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t) \quad (1)$$

where $\Omega(f_t)$ is the regularization term that controls tree complexity:

$$\Omega(f_t) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (2)$$

With:

- T = number of terminal leaves in the tree.
- w_j = weight score on leaf j .
- γ and λ = regularization parameters to prevent overfitting.

2.2 Least Angle Regression (LARS) Scratch

Least Angle Regression (LARS) is a stepwise regression algorithm suited for high-dimensional data [3]. LARS operates equangularly: at each step, it identifies the feature with the highest correlation with the current residual, and moves in the direction of the equangular path (equal angle) among all active features [4, 5, 6].

The mathematical steps of the implemented LARS Scratch algorithm are detailed as follows [5, 7]:

1. **Initialization:** Set the initial prediction $\hat{y}_0 = 0$, index iteration $k = 0$, and active feature set $A = \emptyset$. The target y is centered as $y_{centered} = y - \bar{y}$, and the feature matrix X is standardized to have a mean of 0 and a variance of 1.
2. **Calculate the initial residual:**

$$r_k = y_{centered} - \hat{y}_k \quad (3)$$

where r_k is the residual at iteration k (for the first step, $r_0 = y_{centered}$).

3. **Calculate partial correlation:** Find the correlation vector between the independent features and the current residual:

$$c_j = X_j^T r_k \quad (4)$$

4. **Find maximum correlation:** Identify the maximum absolute correlation among all candidate features:

$$\hat{c} = \max_j |c_j| \quad (5)$$

5. **Update the active feature set:** Add the feature index with correlation $\hat{c}$ to the active set A .

6. **Estimate parameter direction using OLS:** Project the target onto the active feature subspace using Ordinary Least Squares (OLS). To ensure numerical stability under high multicollinearity, the Moore-Penrose pseudo-inverse is utilized:

$$\bar{\beta}_{k+1} = (X_A^T X_A)^+ X_A^T y_{centered} \quad (6)$$

where X_A represents the active feature matrix at the current step.

7. **Determine the optimum step length:** Calculate the distance step size $\hat{\gamma}$ to update regression coefficients:

$$\hat{\gamma} = \min_{j \in A^c} \left\{ \frac{\hat{c} - c_j}{\hat{c} - a_j}, \frac{\hat{c} + c_j}{\hat{c} + a_j} \right\} \quad (7)$$

where a_j is the projection direction component.

8. **Update regression parameters:**

$$\hat{\beta}_{k+1} = \hat{\beta}_k + \hat{\gamma}(\bar{\beta}_{k+1} - \hat{\beta}_k) \quad (8)$$

9. **Update prediction:** Generate the updated active prediction:

$$\hat{y}_{k+1} = X_A \hat{\beta}_{k+1} \quad (9)$$

10. **Iterate:** Repeat steps from step 2 with the new residual r_{k+1} until convergence is achieved ($r \approx 0$) or the maximum active feature limit is reached.

The key advantages of LARS are its high computational efficiency and effectiveness [5]. The model can also be implemented using Python's `scikit-learn` library.

2.3 Evaluation Metrics

Model performance is measured using several statistical metrics to validate prediction accuracy:

- **Mean Absolute Error (MAE):** Average of absolute differences between actual and predicted values.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (10)$$

- **Mean Squared Error (MSE):** Average of squared differences between actual and predicted values to penalize larger errors [8].

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (11)$$

- **Root Mean Squared Error (RMSE):** Root of the MSE to scale errors back to target dimensions.

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (12)$$

- **Coefficient of Determination (R^2):** Measures model fit as the proportion of target variance explained by features [9], describing how much variance of the dependent variable is explained by independent variables [10]:

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (13)$$

or alternatively using squared residuals:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (14)$$

- **Mean Absolute Percentage Error (MAPE):** Average percentage of absolute errors relative to actual target values.

$$\text{MAPE} = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (15)$$

3 Business Reasoning

This chapter explains the business reasoning behind each of the four project goals defined in Chapter 1. Rather than simply listing objectives, it contextualizes why each goal was selected, what operational problem it directly addresses, and what outcome was hypothesized before the analysis was carried out. This rationale-first approach aligns analytical effort with measurable business value.

3.1 Data Engineering with DuckDB

The De Gouden Tarwe bakery chain operates many branches across the Netherlands. Each branch creates daily transaction records, which cover production amounts, sales volumes, waste logs, product prices, and branch details. These records are saved in different database tables. They must be joined, cleaned, and improved before anyone can analyze them or build forecasting models. Choosing DuckDB as the main data engine, instead of common tools like Pandas or SQLite, is based on three practical business reasons [11]:

- **Performance at scale:** DuckDB is an in-process analytical (OLAP) database. It is designed to run fast queries on column-based data. For a dataset with many years of daily sales across dozens of branches, DuckDB runs joins and groupings much faster than row-based databases. This reduces the time it takes to process the data.
- **SQL familiarity for business teams:** Standard SQL is used for all data transformations. This makes the data process easy to read and check by data analysts and store managers, even if they do not know Python. This makes it easier to update the system in the future.
- **SQL familiarity for business teams:** Creating rolling window averages and lag features, which are critical inputs for any demand forecasting model, requires ordered and window-based calculations. DuckDB has built-in support for window functions like `OVER (PARTITION BY ... ORDER BY ...)`. This makes the code shorter and less likely to have errors compared to using Pandas rolling operations.

3.2 Waste Root Cause Analysis and Branch Mapping

Before investing in any predictive modelling systems, it is essential to understand where and why food waste is happening. A data-driven root cause analysis of waste serves two main business purposes. First, it checks if the waste problem is the same across all stores or if it is concentrated in specific branches or product categories. Second, it helps find out if overproduction, which can be fixed with better forecasting, is the main cause of waste, or if other issues like physical damage, quality problems, or customer returns are the main drivers [12]. This difference is very important for deciding how to spend resources. If overproduction causes most of the waste, then a machine learning forecasting model is the right solution. However, if physical damage or poor quality control is the main issue, the company should invest in better handling methods, refrigeration, or supplier quality

checks instead of forecasting models. Therefore, this analysis acts as a strategic checkpoint before building a full machine learning system. Mapping the data at the branch level adds another benefit. It shows which locations have the most waste, allowing the business to focus its audits and test new solutions in those high-loss stores before making changes across the entire network.

3.3 ROI Estimation via Machine Learning

Building a forecasting model is only justified if it delivers quantifiable financial returns. A technical improvement in prediction accuracy say, a 2% reduction in MAE is meaningless to a bakery chain CFO unless it can be translated into euros saved per year. This goal therefore closes the loop between data science output and business decision-making: it converts model performance metrics into projected savings under plausible waste reduction scenarios. The ROI framework also serves a communication function. When presenting the project to branch managers or executive stakeholders who are not data scientists, framing results as 'annual raw material savings of €680,963' is far more compelling than 'an R^2 of 0.52'. By anchoring the business case in financial outcomes, the project justifies operational change and investment in the forecasting infrastructure. Two machine learning models were developed for this goal LARS (implemented from scratch using NumPy) and XGBoost to enable a direct comparison under identical training and testing conditions. This ensures that the model selected for ROI estimation is empirically validated rather than chosen by assumption.

3.4 Production Sales Correlation Analysis

The correlation analysis between planned production, actual production, sales quantities, and daily waste ratios serves as a foundational diagnostic. It answers the question: to what extent does current production planning track actual demand, and where does the system consistently over- or under-shoot? This analysis is particularly important for surfacing the planned vs. actual production gap. In the current manual system, planned quantities are set by branch managers based on experience and intuition. If the data shows that actual production consistently equals planned production (i.e., managers always produce exactly what they planned), it indicates that the planning process itself is the lever to target not execution. Conversely, if actual production frequently deviates from planned quantities, it suggests kitchen-level operational variability that must be separately addressed. The daily waste ratio defined as wasted units divided by produced units provides a normalized metric for comparing waste efficiency across products and time periods, independent of absolute production volumes. This makes it possible to identify which product categories are structurally more prone to overproduction relative to their demand volatility.

4 Operational Waste Exploration & Results

This chapter presents the complete empirical results of the project across all four goals. Each section covers the analytical findings, contextual interpretation, and where applicable an evaluation of the hypotheses formulated in Chapter 3. The results are organized to progress from descriptive waste analysis through predictive modeling outcomes and concluding financial impact projections.

4.1 Data Engineering Pipeline (DuckDB)

Before the analytical and modeling results could be produced, the raw relational database tables were processed through a DuckDB-based feature engineering pipeline. The pipeline performed the following operations:

1. **Table joins:** The waste log, daily production, branch master, and product master tables were joined on `branch_id` and `product_id` keys using SQL `INNER JOIN` operations. DuckDB’s columnar execution engine processed these multi-table joins with minimal latency.
2. **Data cleaning:** Null values in quantity fields were handled by filtering records where produced quantity was zero or negative. Date fields were cast to the `DATE` type and validated for chronological consistency.
3. **Time-series feature engineering:** Three lag and rolling window features were computed per product-branch-date combination using SQL window functions:
 - `lag_1_sold` — the previous day’s sold quantity;
 - `rolling_7_avg_sold` — the 7-day rolling average of sold units; and
 - `day_of_week` — an integer encoding of the weekday to capture weekly demand cycles.

The resulting feature table served as the input dataset for both the LARS Scratch and XGBoost models. This outcome confirms the hypothesis that DuckDB provides a performant and readable platform for time-series feature engineering in a multi-branch retail context.

4.2 Analysis of Primary Causes for Product Waste

Product waste across the De Gouden Tarwe network is recorded under five operational categories. Table 1 presents the total quantity of wasted products and the direct cost of goods sold (COGS) loss associated with each category.

4.2.1 Findings and Discussion

The data reveals a striking and operationally significant finding: waste losses are distributed remarkably evenly across all five categories, with each contributing between €676,450 and €687,798 a range of only €11,348 across all categories. This near-uniform

Table 1: Analysis of Waste Quantity and Cost by Waste Reason

Waste Reason	Total Wasted Qty (Units)	Total Loss (€)
Beschadigd (Physical Damage)	513,364	687,798.09
Verlopen (Expired)	512,335	683,439.48
Kwaliteitscontrole (QC Failure)	511,899	679,396.61
Retour Klant (Customer Return)	513,701	677,733.68
Overproductie (Overproduction)	512,942	676,450.17
Total across All Categories	2,564,241	3,404,818.03

distribution challenges the initial hypothesis that overproduction and expiry would dominate the loss profile.

Instead, the pattern suggests a systemic operational issue rather than a failure concentrated in any single waste category. Every branch, across every failure mode whether physical handling damage, quality control rejection, customer returns, or overproduction appears to contribute losses at approximately the same rate. This uniformity implies that root causes may lie in shared infrastructure factors such as:

- **Centralized kitchen supply chain:** Products are prepared at a central kitchen and distributed to branches. Any systematic overestimation at the sourcing stage would generate waste uniformly across all downstream categories.
- **Standardized disposal logging:** If branch staff categorize disposal events with limited precision, the even distribution may partially reflect classification noise rather than true operational cause diversity.
- **Multi-cause overproduction:** A product produced in excess may be logged as ‘expired’ if it reaches end-of-day unsold, or as ‘QC failure’ if inspectors are strict—effectively converting overproduction losses into other nominal categories.

Despite this even distribution, *Overproductie* and *Verlopen* together account for €1,359,889.65—approximately 40% of total losses. These two categories represent waste that is directly preventable through accurate demand forecasting, as they arise from the demand-supply gap that ML models are specifically designed to close. The remaining 60% of losses (damage, QC, returns) are less addressable through forecasting alone and would require complementary interventions in handling, logistics, and customer management.

4.3 Mapping Branches with the Largest Financial Losses

The De Gouden Tarwe bakery chain operates branches across numerous cities in the Netherlands. Table 2 identifies the ten branches generating the highest cumulative food waste losses over the analysis period.

4.3.1 Findings and Discussion

The branch mapping results reveal two important characteristics of the loss distribution across the network.

Table 2: Top 10 Bakery Branches by Financial Waste Loss

No	Branch Name	Total Wasted Qty	Total Loss (€)
1	De Gouden Tarwe Maastricht	75,016	100,277.00
2	De Gouden Tarwe Delft	73,831	99,701.35
3	De Gouden Tarwe Eindhoven 1	73,971	99,277.85
4	De Gouden Tarwe Arnhem 2	72,678	99,093.52
5	De Gouden Tarwe Haarlem	73,767	98,939.08
6	De Gouden Tarwe Nijmegen	73,861	98,802.75
7	De Gouden Tarwe Leiden 1	73,708	98,775.70
8	De Gouden Tarwe Rotterdam 3	73,999	98,575.11
9	De Gouden Tarwe Leeuwarden	74,231	98,494.45
10	De Gouden Tarwe Utrecht 2	73,241	98,312.15

First, the Maastricht and Delft branches lead the list, each approaching €100,000 in cumulative losses with Maastricht recording the single highest total at €100,277.00. These branches warrant priority operational audits focused on production supply targets shipped from the central kitchen.

Second, and perhaps more noteworthy, is the narrow financial range across all ten top branches: from €98,312 (Utrecht 2) to €100,277 (Maastricht), a difference of less than €2,000. This near-uniform distribution across branches mirrors the uniform distribution across waste categories observed in Section 4.2, further reinforcing the systemic-issue interpretation.

This finding partially confirms the geographic concentration hypothesis the top-loss branches do include larger urban centers but it contradicts the expected magnitude difference. A more differentiated loss distribution would have been expected if individual branch management quality or local demand volatility were the primary drivers. The uniformity instead suggests that the central kitchen planning process, which determines how much product is shipped to each branch daily, is the dominant source of overproduction risk across the entire network.

Operationally, this means that a single centralized forecasting model rather than branch-specific models may be sufficient as a first intervention. A network-level demand forecasting system that improves central kitchen production planning could benefit all branches simultaneously, rather than requiring bespoke solutions per location.

4.4 Machine Learning Model Development and Comparison

To generate demand forecasts that can drive production planning decisions, two machine learning models were developed and evaluated under identical training and testing conditions: Least Angle Regression (LARS), implemented from scratch using NumPy, and XGBoost Regressor, using the standard library implementation. Both models were trained on historical transaction data and validated on a held-out test set to ensure an unbiased performance comparison.

Machine learning is essential for this forecasting task because daily demand for bakery products is influenced by multiple non-linear factors simultaneously day-of-week effects,

weekly sales trends, seasonal patterns, and historical sales trajectories which cannot be captured reliably by simple averaging or manual estimation. With dozens of branches and hundreds of product SKUs, manual production planning is both impractical and systematically biased toward overproduction, as managers prefer the risk of waste over the reputational cost of stockouts.

4.4.1 Model Performance Results

Table 3: Model Comparison: LARS Scratch (NumPy) vs. XGBoost Regressor

Evaluation Metric	LARS	XGBoost Regressor	Winner
Training Time (seconds)	2.08s	3.40s	LARS (39% Faster)
Mean Absolute Error (MAE)	29.20	29.38	LARS (by 0.18 units)
Root Mean Squared Error (RMSE)	41.45	42.63	LARS (by 1.18 units)
R-squared (R^2)	51.92%	49.15%	LARS (by 2.77%)
MAPE (%)	71.13%	71.43%	LARS (by 0.30%)

4.4.2 Findings and Discussion

The evaluation results present a counter-intuitive outcome that directly contradicts the model performance hypothesis stated in Chapter 3. Rather than XGBoost outperforming LARS, the custom LARS implementation achieved superior results across every evaluation metric:

- **Accuracy (MAE):** LARS achieved a MAE of 29.20 units versus XGBoost’s 29.38 units a marginal but consistent advantage of 0.18 units per daily forecast. Over a network with dozens of branches and hundreds of product lines, this marginal accuracy gain translates into meaningfully less systematic overproduction.
- **Error magnitude (RMSE):** LARS recorded an RMSE of 41.45 units compared to XGBoost’s 42.63 units, a difference of 1.18 units. The lower RMSE indicates that LARS produces fewer large forecasting errors particularly important for high-volume products where a single large overproduction event generates significant waste.
- **Explanatory power (R^2):** LARS achieved an R^2 of 51.92% versus XGBoost’s 49.15%. While both models explain approximately half of demand variance a realistic ceiling given the inherent unpredictability of daily retail demand LARS’s 2.77 percentage point advantage indicates a meaningfully stronger fit to the underlying data structure.
- **Percentage error (MAPE):** LARS achieved a MAPE of 71.13% versus 71.43% for XGBoost. The high absolute MAPE values for both models reflect the well-documented difficulty of forecasting fresh food demand at the daily SKU-branch level, where demand volatility is inherently high.

- **Training efficiency:** LARS required only 2.08 seconds to train 39% faster than XGBoost’s 3.40 seconds. For a production system that must retrain daily on new transaction data, this speed advantage reduces operational compute costs and pipeline latency.

The superior performance of LARS on this dataset is consistent with its theoretical strengths: it is particularly effective when the number of active predictors is relatively small and the relationship between features and the target is approximately linear both conditions that apply to bakery demand forecasting, where a limited set of lag and day-of-week features drive most of the predictable signal. Based on these results, the LARS Scratch model was selected as the final production forecasting model. The serialized model binary (`lars_model.pkl`) is the recommended artifact for integration into the branch Point of Sale (POS) system.

4.5 Savings Simulation and Business ROI

Food waste represents a significant source of operational loss in the bakery industry because products that remain unsold still incur the full cost of raw materials and kitchen labor. Improving forecasting accuracy can generate substantial financial benefits by reducing excess production and minimizing waste volume.

Based on the baseline annual food waste cost of €3,404,818.03, a savings simulation was conducted to estimate the projected financial impact of deploying the LARS forecasting system under three plausible waste reduction scenarios (Table 4).

Table 4: Projected Annual Savings by Waste Reduction Scenario

Reduction Scenario	Annual Savings (€)	Annual Savings (IDR)	Assessment
10% Waste Reduction	340,481.80	~IDR 5.9 Billion	Conservative
20% Waste Reduction	680,963.61	~IDR 11.9 Billion	Most Realistic
30% Waste Reduction	1,021,445.41	~IDR 17.8 Billion	Optimistic

4.5.1 Findings and Discussion

The 20% waste reduction scenario is assessed as the most realistic and achievable target for the initial deployment phase. This assessment is grounded in the LARS model’s forecasting performance: with a MAE of 29.20 units per daily forecast, the model brings production decisions significantly closer to actual demand particularly for the *Overproductie* and *Verloopen* categories, which together represent the 40% of losses most directly attributable to demand-supply misalignment.

At a 20% reduction, the projected annual savings of €680,963.61 represent a substantial return that can be realized with minimal operational change. The LARS model can

be integrated into existing POS or ERP systems as a daily batch inference job, requiring no changes to kitchen operations only to the production quantity targets supplied to branch kitchens each morning.

The 10% scenario represents a conservative floor: even if model adoption is partial or branch compliance with AI-generated targets is inconsistent, a 10% reduction is achievable as long as the system is used for at least the highest-loss branches. Conversely, the 30% scenario is achievable in the medium term once the model is retrained on post-deployment data and branch managers build confidence in AI-assisted planning a trajectory supported by evidence from comparable grocery and food retail ML deployments.

This ROI analysis confirms that machine learning does not merely improve forecasting accuracy in the abstract; it delivers measurable, quantifiable financial value. By reducing overproduction, minimizing food waste, and optimizing raw material utilization, the proposed system contributes directly to gross margin improvement and provides a positive return on investment within the first operational year.

4.6 Correlation Analysis: Production vs. Sales and Waste

The correlation analysis examines the relationship between planned production quantities, actual production, units sold, and daily waste volumes across a sample of product-date combinations. This provides a direct diagnostic of how well the current manual planning process tracks actual customer demand.

Table 5: Sample Correlation of Production Chain and Daily Waste Ratio

Date	Product Name	Planned	Produced	Sold	Wasted	Waste Ratio
2019-05-04	Bruin Casino Brood	114	114	105	9	7.89%
2020-11-30	Saucijzenbroodje	91	91	84	7	7.69%
2019-03-17	Fris (cola/fanta/sinas)	92	92	85	7	7.61%
2022-02-20	Tompouce	120	120	111	9	7.50%
2019-12-16	Oliebol Naturel	494	494	457	37	7.49%

4.6.1 Findings and Discussion

The sample data reveals a structurally important pattern: in every recorded instance, planned production equals actual production (Planned = Produced). This confirms the planning compliance hypothesis from Chapter 3 branch kitchens execute precisely what is planned, without deviation. This finding is analytically significant because it establishes that the forecasting intervention must target the planning stage, not kitchen execution.

The waste ratios observed in the sample range from 7.49% to 7.89%, representing a tight cluster of overproduction rates across diverse product categories from standard bread (*Bruin Casino Brood*) to seasonal items (*Oliebol Naturel*) and beverages (*Fris*). Several observations emerge from this data:

- **Consistent overproduction pattern:** Every product in the sample is overproduced relative to demand. No instances of undersupply (sold = produced) appear in the high-waste-ratio tail of the distribution. This aligns with the known managerial tendency to overproduce as insurance against stockouts.
- **Oliebol Naturel, highest absolute waste:** With 37 units wasted on a single day (December 16, 2019), the *Oliebol Naturel* entry illustrates how seasonal or promotional items which have historically high demand but volatile sales carry elevated overproduction risk. A forecasting model that incorporates seasonal patterns from prior years would be especially effective for such items.
- **Tompouce and Bruin Casino Brood consistent 7.5–7.9% waste rate:** The near-identical waste ratios across fundamentally different product types (a pastry versus a bread loaf) further support the systemic-cause interpretation: overproduction appears to be a uniform planning bias rather than a product-specific phenomenon.

From a modeling perspective, the finding that Planned = Produced simplifies the ML pipeline: there is no need to model the kitchen execution gap separately. The forecasting model needs only to generate an accurate daily sold quantity estimate, which can then directly replace the current planned quantity as the production target. This clean mapping from model output to operational action is a key advantage for deployment.

5 Conclusion and Recommendations

5.1 Summary of Findings

This project successfully demonstrates how daily operational waste data from the De Gouden Tarwe bakery network can be transformed into actionable, million-euro saving opportunities through a combination of DuckDB-powered SQL analytics, statistical waste analysis, and comparative machine learning. The key findings across all four goals are summarized below:

1. **Systemic waste distribution:** Food waste losses totaling €3,404,818.03 are distributed remarkably evenly across all five waste categories physical damage, expiry, QC failure, customer returns, and overproduction suggesting a network-wide systemic planning issue rather than isolated operational failures. Overproduction and expiry together account for approximately 40% of total losses, representing the segment most directly addressable by forecasting.
2. **Geographic uniformity:** The top 10 branches by waste loss exhibit a very narrow financial range (€98,312 to €100,277), indicating that no single branch is an outlier. This points to the central kitchen production planning process as the primary lever for network-wide improvement, rather than branch-specific interventions.
3. **LARS outperforms XGBoost:** Contrary to the initial hypothesis, the custom LARS implementation achieved superior performance across all five evaluation metrics, recording a MAE of 29.20 units, RMSE of 41.45, R^2 of 51.92%, MAPE of 71.13%, and training time of just 2.08 seconds, 39% faster than XGBoost.
4. **Strong ROI case:** A 20% waste reduction, the most realistic scenario based on model performance, is projected to generate annual savings of €680,963.61 in raw material costs, with minimal operational change required for implementation.
5. **Planning is the intervention point:** The correlation analysis confirms that actual production consistently equals planned production (Planned = Produced), establishing that the forecasting model output can be directly substituted as the new production target a clean and low-friction deployment path.

5.2 Business Recommendations

1. **Deploy the LARS Scratch Model to POS Systems:** Given that LARS achieves the lowest MAE (29.20 units), trains in 2.08 seconds, and produces predictions directly actionable as daily production targets, the serialized model binary (`lars_model.pkl`) should be integrated into branch Point of Sale or ERP systems to provide automated daily production quantity recommendations. Initial deployment should be piloted at the highest-loss branches, Maastricht and Delft before a network-wide rollout.
2. **Target Central Kitchen Planning, Not Branch Operations:** Since Planned = Produced across all sampled records, the intervention must occur at the central planning stage, not at branch kitchen level. The forecasting model should replace

or augment the current manual estimation process used to determine how much product the central kitchen ships to each branch daily.

3. **Focus Cost-Reduction Efforts on Overproduction and Expiry:** While all waste categories contribute equally to total losses, *Overproductie* and *Verlopen* are the only two categories directly reducible through forecasting. Damage, QC failures, and customer returns require separate operational interventions (handling training, refrigeration improvements, return policy redesign) and should be addressed through parallel workstreams.
4. **Conduct Product-Level Waste Audits for Seasonal Items:** High-volume seasonal products such as *Oliebol Naturel* exhibit high absolute waste volumes. A supplementary analysis of seasonal demand patterns incorporating year-over-year holiday sales data should be used to calibrate production targets for these items more precisely than the general model allows.
5. **Retrain the Model Quarterly:** Consumer demand patterns in food retail shift with seasons, economic conditions, and local events. To maintain forecasting accuracy, the LARS model should be retrained on rolling 12-month windows of transaction data at minimum on a quarterly basis, with monthly retraining preferred if compute resources allow.
6. **Establish a Waste KPI Dashboard:** Operational adoption of the forecasting system should be monitored through a real-time waste KPI dashboard tracking daily waste ratio per branch, category-level loss trends, and forecast accuracy (MAPE) over time. This provides management with ongoing visibility into ROI realization and early signals of model degradation.

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